

CLASS XII · LECTURE 6

Arrays

Option I · One to two periods · Needed for lists, totals and for questions that cross over from the programming syllabus

By the end of this lecture you can

- Declare, initialise and index a one-dimensional array.
- Find the total and the average with a loop.
- State that the first index is 0 and the last is size minus one.

One name, many values of the same type**DEFINITION Array**

An array is a collection of values of the same type, stored one after another under one name. Each value is an element. An element is reached by an index.

`int marks[5];` creates five integers: `marks[0]` through `marks[4]`. There is no `marks[5]`. Using that index steps off the end of the array. This is the error to mention even if you are not asked, because it shows you understand the size. You may initialise in the declaration: `int marks[5] = {70, 81, 64, 90, 55};`.

```
#include <stdio.h>

int main(void) {
    int marks[5];
    int i, sum = 0;
    for (i = 0; i < 5; i++) {
        printf("Enter marks: ");
        scanf("%d", &marks[i]);
        sum = sum + marks[i];
    }
    printf("Total = %d\n", sum);
    printf("Average = %f\n", sum / 5.0);
    return 0;
}
```

Program 6.1 Total and average of five marks. The loop index is the array index.

The address operator is on `marks[i]`, not on the whole array name, because `scanf` needs the address of the element being read. Dividing by `5.0` forces a fractional average. Dividing by the integer 5 would chop the fraction.

A two-dimensional array is a table: `int grid[3][3];` has three rows and three columns, indexed from 0. You need two nested loops, the outer for the row and the inner for the column. Paper II's C option rarely asks for a matrix. Know the declaration and the idea that it is a table of one type. Do not start using C++ `string` or `cin` here. A short text is a character array, `char name[20];`, read with a width-limited `scanf`.

EXAM TIP Say the type of array in the first line

If a question says "explain an array and its types", write: one-dimensional (a list) and two-dimensional (a table). Give one declaration of each. Then one loop for the list. Stop.

Checkpoint

Try the questions before you read the model answers. Section B answers on the real paper should stay within about six or seven lines.

Multiple choice

1. In `int a[5]`, the last legal index is:

- A) 5
- B) 4
- C) 6
- D) 1

Short answers

S1. Declare an array of 5 integers and write a loop that reads values into it.

Model answers

Cover this section while you attempt the checkpoint. The wording is a full-mark style answer, not the only acceptable wording.

MCQ 1. B. Indexes run from 0 to size – 1.

S1. `int marks[5];` then for (`i = 0; i < 5; i++`) `scanf("%d", &marks[i]);`. The first element is `marks[0]` and the last is `marks[4]`. The `&` is required because `scanf` needs the address of each element.